worked for the company between 2 and 4 years and have salary between 40K and 80K... and give them a raise[14]!

This search translates into a rectangle with boundaries parallel to the dataset feature axes, meaning that in our boundaries each feature is independent on any other feature. If, instead, we had conditions that would mix more than one feature (f.i., salary lower than 15K for each year of tenure), then the boundaries of the search region would be segment of generic lines, not parallel to any axis: in that case the problem becomes harder to solve, and we might need something more complex, like the simplex algorithm[15], to search the points satisfying our range query.

What's the performance of both region-searches? Well, as you can imagine, it heavily depends on the regions we search: we go on a full range from two edge cases:

- For very small regions intersecting only a single branch corresponding to a leaf, we will prune every other branch, and just follow a path to the leaf: so the running time will be `O(h)`, where `h` is the height of the tree – `O(log(n))` for a balanced tree with `n` points.
- When the region is large enough to intersect all points, then we will have to traverse the whole tree, and the running time will be `O(n)`.

Listing 9.12 The pointsInRectangle method

```
function pointsInRectangle(node,
rectangle)                              #1
  if node == null
then                                                    #2
```